

4NL3 Project Step 3

Team Members

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Annotation Analysis

Agreement

Cohen's Kappa was the agreement metric chosen for the subset of data that consisted of two annotators in our data. The reason being, this metric is used for pairwise agreement between two annotators, which fits our dataset. This metric produced a value of 0.496. This value means the annotators had moderate agreement. Essentially, annotators had 49.6% better agreement than just guessing labels randomly (random chance). This also means there were many disagreements. How these disagreements were dealt with are discussed in the next section.

Ground Truth

A mix of majority vote and adjudication were used to determine the ground truth label for each book blurb. If all annotators agreed on a single label, then that label was assigned as the ground truth. If annotators disagreed, then a manual review took place. This manual review involved selecting the final label by choosing between the two labels provided by the annotators, based on which one best aligned with the text (like adjudication, but without discussion). In other words, a tiebreaker was selected.

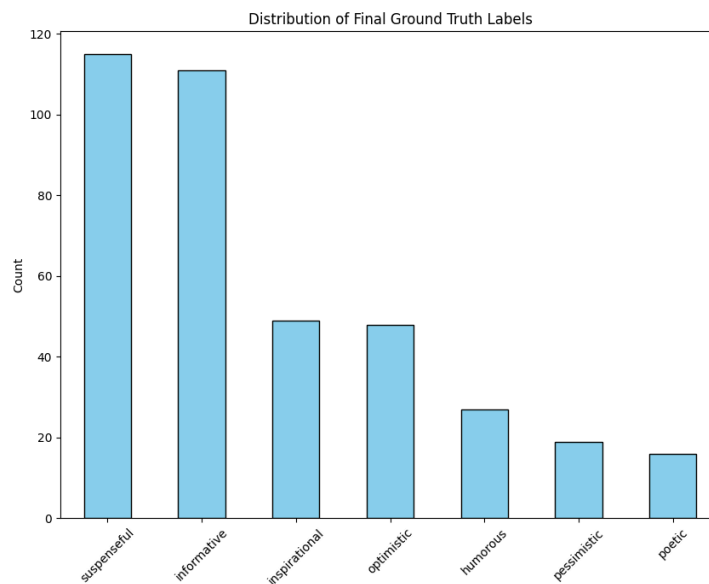
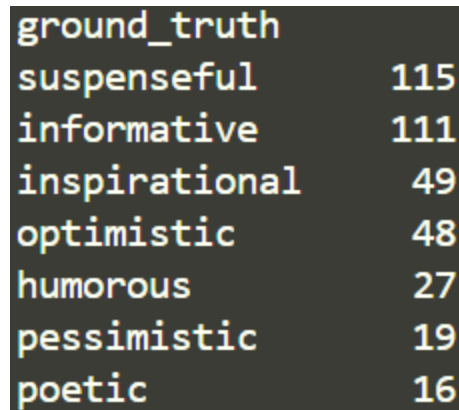
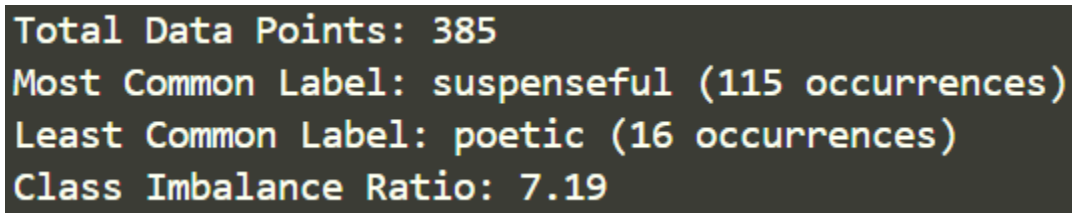


Figure 1. Distribution of final ground truth labels



ground_truth	
suspenseful	115
informative	111
inspirational	49
optimistic	48
humorous	27
pessimistic	19
poetic	16

Figure 2. Label frequency



Total Data Points: 385
Most Common Label: suspenseful (115 occurrences)
Least Common Label: poetic (16 occurrences)
Class Imbalance Ratio: 7.19

Figure 3. Class imbalance ratio

Figure 1 shows the distribution of the final ground truth labels. ‘Suspenseful’ and ‘Informative’ occurred the most, with the other 5 labels appearing less than half as many times as the most frequent labels, as can be seen in Figure 2. Specifically, the most common label - ‘Suspenseful’ - appeared roughly 7.19 times as much as the least common label - ‘Poetic’. This is a fairly large imbalance. This could be due to a number of reasons. Firstly, the nature of the texts themselves could allude to being more ‘Suspenseful’ or ‘Informative’, than ‘Poetic’ or ‘Pessimistic’ (another low-frequency label). Additionally, annotator bias could be a factor. Perhaps annotators feel more comfortable identifying ‘Suspenseful’ texts than they do identifying ‘Poetic’ texts. It may not be as clear when a piece of text is ‘Poetic’ as opposed to other labels. Furthermore, there could be an uneven distribution, and if a different set of texts were used, different results could be achieved. Perhaps clearer ‘Poetic’ examples could’ve been given to annotators, and/or a less niche label could’ve been chosen, like ‘Philosophical’, which is sometimes confused with ‘Poetic’.

The issue with this data is that when training a model, it may overtrain on ‘Suspenseful’ and ‘Informative’ labels, and undertrain on ‘Pessimistic’ and ‘Poetic’ labels, making the model unreliable in certain cases. This will need to be accounted for when training the model.

One interesting thing to note is, many manual reviews consisted of annotators disagreeing when it came to identifying a text as ‘Informative’ or ‘Inspirational’. This was the case when it came to biographies. When conducting the manual review, the tone and use of words was carefully considered to deduce whether the text was more trying to motivate and inspire the reader, or to

inform them of something. On our part, we should've given a clear-cut way to select between the two labels.

Baseline Models (generated data from train.csv and val.csv only, no predictions made for hidden test.csv)

Simple Baselines

The evaluation of baseline models shows that a **random baseline**, which assigns labels randomly, achieves an accuracy of 15.52%, aligning with the expected probability of 1/7 ($\approx 14.3\%$) given seven possible tone options. Precision, Recall and F1 scores are particularly low for minority labels such as "pessimistic". Labels like "optimistic" and "inspirational" had 0.00 in both precision and recall scores, indicating that the random label assignment was never correct.

Classification Report:				
	precision	recall	f1-score	support
optimistic	0.00	0.00	0.00	4
informative	0.50	0.18	0.26	17
suspenseful	0.18	0.25	0.21	8
inspirational	0.00	0.00	0.00	7
poetic	0.14	0.33	0.20	3
pessimistic	0.11	0.50	0.18	2
humorous	0.20	0.12	0.15	17
accuracy			0.16	58
macro avg	0.16	0.20	0.14	58
weighted avg	0.24	0.16	0.17	58

Figure 4. Classification Report for the Random Baseline Model

Using only the train.csv for Codebench. The most common tone in the dataset is "informative". A **majority baseline**, which always predicts "informative," performs significantly better, reaching 29.31% accuracy. This is to be expected since in Annotation Analysis we have already found that there are significantly more data points labeled as "informative". This also highlights a strong class imbalance which potentially skews model predictions and limits accuracy for less frequent tones. The figure below shows the "informative" label to have 100% recall which is to be expected since all data points were labeled as "informative". Similarly, Precision is around 30% because data points labeled "informative" account for around 30% of total data points and the rest are false positives.

Classification Report:				
	precision	recall	f1-score	support
optimistic	0.00	0.00	0.00	4
informative	0.29	1.00	0.45	17
suspenseful	0.00	0.00	0.00	8
inspirational	0.00	0.00	0.00	7
poetic	0.00	0.00	0.00	3
pessimistic	0.00	0.00	0.00	2
humorous	0.00	0.00	0.00	17
accuracy			0.29	58
macro avg	0.04	0.14	0.06	58
weighted avg	0.09	0.29	0.13	58

Figure 4. Classification Report for the Majority Baseline Model

Trained Model

A **Random Forest classifier** was selected due to its effectiveness in multi-class classification and ease of implementation. It is trained on TF-IDF vectorized text data. To address class imbalance, the model uses class weighting to give more importance to underrepresented tones. With 300 decision trees (n_estimators=300), the model improves accuracy by reducing variance, making it more robust although computation time is slightly longer. Additionally, a fixed random state (random_state=42) ensures reproducibility. The model is first trained on train.csv then validated on val.csv. The model has validation accuracy of 51.72%.

Classification Report:				
	precision	recall	f1-score	support
humorous	0.00	0.00	0.00	4
informative	0.45	0.82	0.58	17
inspirational	0.00	0.00	0.00	8
optimistic	0.00	0.00	0.00	7
pessimistic	0.00	0.00	0.00	3
poetic	0.00	0.00	0.00	2
suspenseful	0.59	0.94	0.73	17
accuracy			0.52	58
macro avg	0.15	0.25	0.19	58
weighted avg	0.31	0.52	0.38	58

Figure 5. Classification Report for the Random Forest Model on validation dataset

Although the accuracy has improved in comparison to both simple baseline models. There is little improvement on the classification of minority labels. The Random Forest model exhibits severe class imbalance, performing well only on the "informative" (45% precision, 82% recall) and "suspenseful" (59% precision, 94% recall) classes, while completely failing to classify other tones such as "humorous," "inspirational," "optimistic," "pessimistic," and "poetic," all of which have an F1-score of 0.00. This indicates that the model is heavily biased toward the majority classes, likely due to the uneven distribution in the training data. As a result, the model strongly favors "suspenseful" and "informative," while largely ignoring the minority classes, leading to low precision and recall for less frequent labels. The macro F1-score of 0.15 further confirms that performance is poor across all categories, while the weighted F1-score of 0.31 suggests that predictions are disproportionately skewed toward high-frequency classes. This imbalance significantly limits the model's ability to generalize effectively to underrepresented tones.

Codabench Link: <https://www.codabench.org/competitions/6675/>

PowerPoint Link:

https://docs.google.com/presentation/d/1eot6R00g0Jx_nnvVRTz6t-ooU_Ps0RXS/edit#slide=id.p1